RV College of Engineering®

IDEA

Industrial Engineering Association
RV College of Engineering

DEPARTMENT OF INDUSTRIAL ENGINEERING AND MANAGEMENT

THE INSIGHT

A NEWSLETTER INITIATIVE FROM IDEA

2026

JANUARY-FEBRUARY

Edition

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JANUARY - FEBRUARY 2026 EDITION

Editor's Note


Dr. Rajeswara Rao KVS
Head of Department
IEM, RVCE


Dr. Ramaa A
Associate Dean - P&T,
Associate Professor,
IDEA Secretary, Dept. of IEM, RVCE

Welcome to the IEM Department Newsletter!

Stay informed and connected with the latest updates on departmental events, research breakthroughs, student achievements, and faculty highlights. From academic milestones to innovations in teaching and learning, we bring you stories that reflect the vibrant life of the Industrial Engineering and Management community.

This newsletter also showcases extracurricular activities, including hackathons, technical fests, cultural events, and student club initiatives. We aim to foster collaboration, creativity, and leadership among students while providing a platform to explore interests beyond the classroom.

We hope this newsletter leaves you feeling informed, connected, and proud.

Warmly,
The Editorial Team.


Rakshitha R


Manoj Bhat


Dheeraj Aradhya M

FACULTY THOUGHTSPACE

The Next Frontier of Industrial Engineering

Industrial Engineering is evolving rapidly due to technological advancements, globalization, and the growing need for efficient and sustainable systems. Traditionally focused on improving manufacturing productivity, Industrial Engineering is now expanding into digital, service, and knowledge-based industries. The future of the discipline lies in integrating technology, data analytics, and human-centered design to optimize complex systems.

Chronological Evolution: From 4.0 to 5.0: To orient IE Professionals, it is essential to frame where they are standing in the timeline of industrial history.

- Industry 4.0 (The Era of Data): Focused on connectivity, IIoT, and "Dark Factories" (fully automated). The primary goal was Efficiency & Speed.
- Industry 5.0 (The Era of Purpose): The 2026 shift toward Human-Centricity, Sustainability, and Resilience. The goal is no longer just "output," but "well-being and planetary health."
- Cause & Effect: The hyper-optimization of 4.0 created "brittle" supply chains. The "effect" is the 2026 focus on Resilience Engineering—designing systems that can absorb shocks rather than just lean them out.

Research Frontiers: The 2026 Edge: These are the specific areas where IE research is currently meeting industrial applications

1. Quantum Operations Research (Q-OR): Moving beyond classical linear programming. Quantum algorithms (like QAOA) are now used for "combinatorial explosions" in global logistics that classical supercomputers can't solve in real-time. The IE Professionals need to understand Hybrid Quantum-Classical Workflows—using classical AI to manage data and Quantum processors for the "hard" optimization.
2. Industrial Bio-manufacturing & Circularity: IE is moving into the "Bio-foundry." Using synthetic biology and CRISPR to "grow" materials rather than extract them. The IE Role could be designing the scale-up of fermentation loops and "Zero-Waste" circular ecosystems where one factory's waste is another's feedstock.
3. Cyber-Physical-Social Systems (CPSS): Expanding beyond CPS (sensors + machines) to include the Social (human behavior/psychology). The Research direction could be using "Digital Twins of the Human" to simulate cognitive load and fatigue, ensuring that AI-driven speed doesn't lead to worker burnout.

The Skill Synthesis: Hard vs. Soft: For IEs to be "Future-Proof," they must marry these two domains:

Technical "Hard" Frontiers	Systemic "Soft" Leadership
Explainable AI (XAI): Moving from "Black Box" models to AI that can explain *why* it suggested a production change.	Ethical Governance: Ensuring AI decisions don't inadvertently discriminate or bypass safety protocols.
Edge Computing: Shifting logic from the cloud to the sensor level for millisecond response times.	Change Fitness: The ability to lead an organization through constant tech-driven pivots without "change fatigue."

Critical Thinking: The Growth vs. Sustainability Paradox: IEs shall think critically by presenting these two contrasting "takes" currently debated in the field:

- Take A (Techno-Optimism): Advanced tech (AI/Quantum) will solve the climate crisis through hyper-efficient resource use.
- Take B (Low-Tech/De-growth): True sustainability requires "Intelligent Restraint"—designing simpler, repairable, local systems (Low-Tech) rather than piling on more energy-intensive technology.

Bridge Table: Mapping Traditional IE to 2026 Frontiers: This table translates traditional IE concepts into the "Future-Proof" competencies required for Industry 5.0.

Traditional IE Course	Traditional Focus	2026 Research Frontier / Shift	The "Future-Proof" Bridge
Operations Research	Linear & Integer Programming; Simplex Method.	Quantum-Inspired Optimization (Q-OR)	Shifting from "Optimal Solutions" to "Real-time Heuristics" for combinatorial explosions in global logistics.
Manufacturing Processes	Tooling, CNC, Casting, and Assembly lines.	Industrial Bio-manufacturing	Moving from subtractive/additive manufacturing to "Cellular Factories"—using biological processes as production lines.

Human Factors & Ergonomics	Physical limits, lifting safety, and workstation layout.	Cyber-Physical-Social Systems (CPSS)	Expanding to Cognitive Ergonomics: Measuring worker mental load and trust in AI-collaborative environments.
Supply Chain Management	"Just-in-Time" (JIT) and Lean methodologies.	Autonomous & Resilient Supply Networks	Moving from "Efficiency-first" to "Resilience-first": Designing systems that can self-heal after geopolitical or climate shocks.
Quality Control	Statistical Process Control (SPC) and Six Sigma.	Predictive Digital Twins	From *reactive* defect detection to *proactive* simulation—correcting the process in the "Virtual Twin" before the physical defect occurs.
Facilities Planning	Physical layout and material handling flows.	Circular Bio-Economy Layouts	Designing for "Closed-Loop" flows: The facility is no longer a linear line but a metabolic system where waste is reintegrated.

By combining strong analytical foundations with emerging digital technologies and sustainability principles, industrial engineers can effectively prepare for the future. Continuous learning and adaptation will enable them to design efficient, intelligent, and resilient systems across manufacturing and service industries.

Further Reading:

1. Maretto, V., Faccio, M., & Battini, D. (2025). Adaptive Social Manufacturing: A Human-Centric, Resilient, and Sustainable Framework. International Journal of Production Research.
2. European Commission (2024/2025 Updates). Industry 5.0: Transitions toward a Sustainable, Human-Centric and Resilient European Industry.
3. World Economic Forum (2026). Global Cybersecurity Outlook 2026: The Resilience Imperative.
4. MDPI Sustainability (2025). Impact of Circular Bio-economy on Industry's Sustainable Performance: A Critical Literature Review.

Dr. C K Nagendra Guptha
Professor
Dept of IEM,RVCE

ALUMNI SPEAKS

How Industrial Engineering Shaped My Career

When students choose Industrial Engineering, they often hear a common question: “What will you actually do with it?” I had the same question when I began my B.S. in Industrial Engineering at RV College of Engineering. Today, I’m a Director, Product Management at Palo Alto Networks, where I lead products in Data Security Platform (UEBA - User & Entity Behavior Analytics, Incident Management) and AI Access Security. The path from Industrial Engineering to AI security may sound unusual—but Industrial Engineering prepared me perfectly for it.

Industrial Engineering Teaches You to Think in Systems :

Industrial Engineering is fundamentally about systems thinking—understanding how technology, people, and processes interact. Instead of focusing on just building a single component,Industrial Engineers ask:

- Where are the inefficiencies?
- What does the data reveal?
- How do we improve the entire system?
- How do we scale solutions to millions of users?

That mindset shaped every stage of my career.

Early Career: Process Optimization at Intel :

My career began at Intel, where I worked on demand planning. Using Industrial Engineering principles like forecasting, process optimization, and operational modeling, I helped improve planning across product lines and identified efficiencies that avoided infrastructure costs. This was my first experience seeing how data + systems thinking can transform operations.

Data-Driven Products at Nintendo:

At Nintendo, I applied Industrial Engineering in a completely different domain—product analytics. I lead product feedback for Nintendo Switch and Pokémon Go, via product telemetry enabling teams to detect issues early and continuously improve the player experience. Industrial Engineering taught me something powerful here: data is the fastest way to improve systems.

Scaling CRM and AI Platforms at Microsoft:

At Microsoft, I worked on enterprise platforms like Microsoft Dynamics 365 and Microsoft Copilot Studio. These systems served millions of users worldwide. My work focused on improving engineering fundamentals, product led growth, optimizing system workflows, and building AI-driven solutions that set product market-fit, drove product adoption and scaled quickly. Industrial Engineering helped me approach product development as a large-scale optimization problem.

Advice for Industrial Engineering Students:

If you're studying Industrial Engineering today, focus on three skills:

- Data & Analytics - Learn Python, statistics, and data visualization.
- Systems Thinking - Always look at the end-to-end workflow.
- AI + Operations - The future lies in combining AI with operational systems.

AI + Industrial Engineering cool project ideas for the Good of the Community:

1. AI System to Reduce Food Waste (literally solve world hunger) :
Build a system that predicts excess food in restaurants or cafeterias and routes it to nearby shelters. Impact: Reduce waste and help communities facing food insecurity.
2. Emergency Response Optimization:
Use traffic data and predictive models to recommend optimal routes for ambulances and emergency vehicles. Impact: Reduce emergency response time and save lives.

3. Hospital Wait-Time Predictor:
Build an AI model that predicts emergency room wait times based on patient inflow and staffing. Impact: Help hospitals allocate staff more efficiently and improve patient care.
4. Smart Public Transport Optimizer:
Use data from buses, trains, and passenger demand to dynamically adjust schedules. Impact: Reduce commute times and carbon emissions.
5. Disaster Resource Allocation:
Design a system that optimizes the distribution of relief supplies during floods, earthquakes, or hurricanes. Impact: Ensure aid reaches the right place faster.
6. Water Usage Optimization:
Create an AI system that monitors water consumption across a city and predicts leaks or waste. Impact: Improve sustainability and conserve water.

Final Thought:

Industrial Engineering gave me one powerful mindset: The most valuable engineers are those who can design and improve entire systems. Whether you are optimizing supply chains, building AI products, or improving community services, Industrial Engineering gives you the tools to solve some of the most important problems in the world. Adding an infographic for problem solving approach

VEENA ANANTCHANDRAN MADHUR
2004-2008
Director, Product Management
at Palo Alto Networks

STUDENT PLACEMENTS HIGHLIGHTS

STUDENT PLACEMENTS HIGHLIGHTS

STUDENT PLACEMENTS HIGHLIGHTS

HACHIDORI

SOLUTIONS ENGINEER
4LPA

Swati

Stackbox

DESIGN INTERN
10LPA

Syed Nadeem

MEMORANDA OF UNDERSTANDING

A Memorandum of Understanding (MoU) was signed between **Cettlx Services Pvt. Ltd**. (VealthX) and **the Fintech Center of Excellence (FCoE)** on 18-02-2026, with **validity until 2030**.
The collaboration focuses on training and software development in the fintech domain, aiming to enhance student skills, foster industry-relevant expertise, and promote innovation through practical exposure and joint initiatives.

STUDENT ACHIEVEMENTS

Saptarshi Mandal, a second-year (third semester) student and member of RVCE's dance team **Enigma**, showcased exceptional talent by securing **1st place at multiple intercollegiate** competitions, including IIIT Bangalore, Mount Carmel College, NMIT, and XIMERA Institute during January–February 2026.
His consistent victories in Western dance highlight his dedication, excellence, and team spirit, bringing great pride to Team Enigma and the institution.

The **RVCE Football Team** secured **1st place at the VTU** event held at SJMIT, Chitradurga, during September 2025, with **Poorvi Girish**, a first-semester student, being part of the winning team.
This achievement reflects strong teamwork, dedication, and competitive spirit, bringing pride and recognition to the institution at the intercollegiate level.

Ritika S Ghattad and **Nischal B D**, third-semester students, represented the institution in **Hockey** at the **South Zone Inter University Championship (National)** held at Sathyabama Institute of Science and Technology, Chennai, in January 2026.
Their participation reflects commendable dedication and sportsmanship, bringing recognition to the college at a prestigious national-level event.

STUDENT-FACULTY PUBLICATIONS

“A Comprehensive Review of Temperature Monitoring and Cold-Chain Integrity Across IoT in Vaccine Storage” by Yashwanth L, Sneha S. Shrigiri, Sinchana Chetan, and Ritika S. Ghattad, published in the International Journal of Emerging Science and Engineering (IJESE), Volume 14, Issue 3 (pp. 10–14) (International).

“A Digital Twin Framework for Dual-Path Estimation of Battery State of Charge and State of Health Using Hybrid Physics-Informed Machine Learning” by Andhe Dharani, Siva Subbarao Patange, M. Krishna, Raja Vidya, R. Bindu, and K. N. Subramanya, published in IEEE Access (International, Scopus Indexed).

“A Comparative Study and Review of Optimization Methods for Strategic Allocation of EV Charging Stations” by Dr. C. K. Nagendra Gupta, Rishabh Patawri, Samadrito Mukherjee, Satyam Kumar published in the Industrial Engineering Journal, Volume XIX, Issue No. 01 (National).

“Effectiveness of Agile Methodologies in Hybrid Project Environments” by Vikram N. Bahadur Desai, published in the International Journal of Scientific Research in Engineering and Management (IJSREM), Volume 09, Issue 12 (International).

“Forecast Accuracy Improvement and Purchase Order Automation Using a Multi-Model Approach in Semiconductor Supply Chain: A Case Study” by J. Gohitha Maheshwari, N. Tanushri, Ramana Gowda, Vatsal Singh Bhutiyal, and K. V. S. Rajeswara Rao, published in IEEE Access, Volume 14 (International, Scopus Indexed).

“Plasmonics-Driven Graphene Electronics for Beyond-5G and 6G Communication Systems” by Dr. Vikram N. B., published in the Journal of Material Science (International).

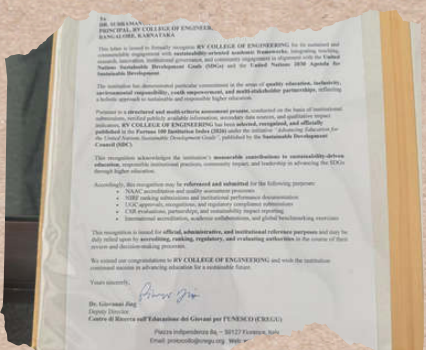
“Sustainable Green Roof Hydroponic Farming Using Hydrogen Induced Water from Green Hydrogen Production: A Comparative Study of Plant Growth, Nutritional Quality, and Predictive Modelling” by Smriti J. Abraham, Sudhanva Patil, Pranathi S., Movva Sai Lalithadevi, Stuthi Shrisha, and Dr. M. N. Vijaykumar, presented at The Second International Conference on Innovative Engineering Design (ICoIED 2026), held on February 16–17, 2026, at IIT Roorkee, Uttarakhand, India (International).

“Psychological Profiling of Digital Dependency Using Machine Learning and Behavioral Clustering Techniques” by C. Bindu Ashwini, C. S. Harsha, B. M. Sreekar, and Harshith K. Murthy, published in IEEE Access, Volume 14 (International, Scopus Indexed).

AWARDS/ACHEIVEMENTS

Dr. Ramaa A received the IUCEE Individual Honorary Membership at ICTIEE 2026, held at GITAM University from 7th to 10th January 2026, in recognition of her contributions to engineering education and academic excellence.

RV College of Engineering has been recognized in the **Fortune 100 Institution Index 2026**, securing the **11th rank under the National Institution Index, published by the Sustainable Development Council,** for its contributions toward advancing education aligned with the United Nations Sustainable Development Goals (SDGs). The recognition was conferred at the **Global Conference on Education for a Sustainable Future (GCE 2026)** held at the **United Nations Conference Centre (UNCC), Bangkok, Thailand**, and **issued by the Sustainable Development Council (SDC) in collaboration with UNESCO.**
Dr. K. N. Subramanya, received the recognition at the Global Conference from **H.E. Yeezang De Thapa, Hon’ble Minister for Education and Skills Development**, **Kingdom of Bhutan, along with Mr. Gokulnath Mathiazhagan, Secretary-General, SDC (India),** acknowledging the institution’s excellence in sustainability-driven education, research, and global impact.



Certificate Of Recognition

RV College of Engineering

SEMINARS - CONFERENCES - WORKSHOPS

20TH INDIA DIGITAL SUMMIT

Attended by : Dr. M N Vijaykumar
Venue : The Leela Bhartiya City, Bengaluru
Date : 29TH - 30TH January 2026
India's AI Moment: Leveraging the Intelligent Economy
Join the next wave of insights, innovation and inspiration.

The Fourth Annual Taguchi Methods Symposium 2026

Attended by :Dr. C K Nagendra Guptha, Dr. Vijaya Kuamr M N
Date: 2nd February 2026
Venue: TVS Institute for Quality Leadership (TVS IQL) in Anekal. organised by TVSM in collaboration with IFQM (Indian Foundation for Quality Management)

One-Day Workshop on Introduction to MIP Modeling in Python and Performance Tuning for Large-Scale Optimization.

Attended by : Dr. Vivekanand S Gogi, Dr. N S Narahari
Venue: Department of Management Studies, Indian Institute of Science, Bengaluru
Date : 28TH February 2026

ICTIEE 2026

Attended by : Dr. Ramaa A,
Dr.K. N. Subramanya
Venue : GITAM University, Bangalore
Date : 7TH to 10TH January 2026

CESS Talk SeriesViksit Bharat Shiksha Adhishtan (VBSA) Bill, 2025: Context, Principles and Policy Considerations.

Attended by : Dr. N S Narahari
Date : 31ST January 2026
Venue : Online

Dr. N. S. Narahari delivered a guest lecture during a **one-day training program** on **Reliability Engineering in association with Zero Defect Consultants, Electronic City Post**, Bengaluru – 560100, on 20th February 2026.
The program focused on enhancing understanding of reliability principles and practices, providing participants with practical insights into improving system performance, quality, and operational efficiency.

Dr. Vijayakumar MN attended the meeting at Mechanical Engineering Department at **JSS Academy of Technical Education** as a member of the **Department Advisory Board (DAB)**, Bengaluru, on 26 February 2026.

Dr. N. S. Narahari attended the event **"Sustainability and Green Engineering – Role of IE Professional"** on the occasion of IIIE Foundation Day, held on 31st January 2026.
The event was **organized and attended by Dr. C. K. Nagendra Gupta and Dr. Ramaa A**, who serve as **Executive Committee Members of IIIE.** The session highlighted the importance of sustainable practices and the evolving role of industrial engineering professionals in promoting environmentally responsible solutions.

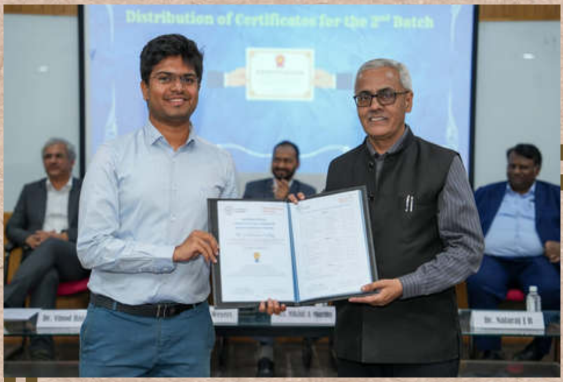
Dr. Vivekanand S. Gogi was invited by the **Department of Industrial Engineering & Management at Ramaiah Institute of Technology** to serve as an **External Member of the Department Advisory Board**, contributing to academic guidance and departmental development.

INTERNATIONAL CERTIFICATE COURSE AT GERMANY

The International Certificate Course on "Smart Manufacturing," jointly offered by RV College of Engineering (RVCE), India, and Fachhochschule Dortmund (FH Dortmund), Germany, was conducted from 23.02.2026 to 13.03.2026 with 26 participants from RVCE and 7 from FH Dortmund.

The inauguration of Batch 3 was held on February 23, marking the continuation of this international academic collaboration. During the event, certificates were also distributed to Batch 2 participants, recognizing their successful completion of the program. Delegates from FH Dortmund also visited the VTU Regional Office, Nagarabhavi, on 24th February 2026 to explore academic and research collaborations.

As part of the program, Prof M. N. Shruthi completed the module on "CNC Turning & Milling Programming – Theory and Hands-on Experience," conducted at the CNC and Robotics Lab, RVCE, on 25th February 2026, gaining practical exposure to advanced manufacturing systems. Prof. Dr. N. S. Narahari, Dr. Ramaa A, Prof. Dr. Vikram N. B., and Prof. Bhaskar M. G. completed the Data Analytics module as part of the same program.



MAJOR EVENTS

Dr. N. S. Narahari served as a panel member on February 17 at **IEDC, DSCE**, for the evaluation of ideas under the National Institutional Startup Funding Scheme.
A total of 34 innovative ideas were pitched by students of DSCE, showcasing creativity and entrepreneurial potential, contributing to the institute's growing startup ecosystem.

Mr. Mahidar Jayaram, a **1998 batch alumnus** and **Director of Wyzmindz**, visited the department and engaged in discussions regarding a **potential MoU with the SCM Centre of Excellence and the department**.
The interaction focused on exploring collaborative opportunities, strengthening industry–academia linkage, and enhancing practical exposure for students.

Turing Award-winning professor **Jeffrey D. Ullman** visited RV College of Engineering during and spoke on the **impact of AI in education.** He emphasized that while AI aids coding, strong fundamentals are essential, and AI should be viewed as a collaborator rather than a replacement.

Dr K. N. Subramanya, participated in the AICTE meeting focused on the **role of Artificial Intelligence in engineering education.** During the session, the importance of integrating AI as a collaborative tool rather than a replacement was highlighted. Emphasis was placed on strengthening fundamental coding skills while leveraging AI to enhance productivity and innovation. The discussion also explored how AI-driven approaches can transform teaching, learning, and future career opportunities for students.

The **26th Board of Studies** meeting for the BE–Industrial Engineering and Management program was held on Friday, 23rd January 2026, , focusing on academic planning and curriculum enhancement.

Dr. Ramaa A visited **Quest Global** on 13/01/2026 for discussions on industry-offered electives, aimed at strengthening industry–academia collaboration and enhancing curriculum relevance.

Dr NS Narahari attended **Silver Jubilee Celebrations** of **RV Institute of Management** as a special invitee at NMKRV Campus, Bengaluru on 21 st February 2026.

Dr NS Narahari was one of the **Panelists** for level 2 scrutiny of shortlisted ideas under **NISP/IEDC Seed funding 2025-2026** at **Dayananda Sagar College of Engineering,** on 17th February 2026.

Dr Rajeswara Rao attended the **BoS Meeting** at **Ramaiah Institute of Technology** on 7th February 2026.

Meghasree V, a **research scholar** from the research center of IEM, completed her **PhD Viva Voce** examination in the area of **"Competitiveness Enablers in Handicraft Toy Manufacturing Industries based on ISM-Fuzzy MICMAC Analysis"** on 20th February 2026 under the **guidance of Dr. C K Nagendra Gutha.**

Srividya Sriraman (6th sem), **Pranav Ramakrishnan (6th sem)**, and **Syed Nadeem (8th sem)** represented **Ashwa Racing, RVCE at Formula Bharat 2026**, held from January 21–26 at **Kari Motor Speedway, Coimbatore.**
They played a key role in the static event – **Business Plan Presentation (BPP)**, where participants were required to develop a comprehensive business plan from scratch based on either the race car or one of its components.
The team **secured 2nd place** in BPP and placed **P6 overall** in the competition.
It was an intense and fast-paced experience, with constant pressure and little time to pause. There were many lessons along the way, shaped just as much by mistakes as by progress. Through it all, the team stayed focused on maintaining the standard and reputation that Ashwa Racing is known for.

FACULTY MEMBERSHIP - PROFESSIONAL BODIES

Dr. C. K. Nagendra Gupta is a member of the Institute of Industrial and Systems Engineers (IISE), USA (Member ID: 900071060)

Dr. Ramaa A is a Member of the Institute of Industrial and Systems Engineers (IISE), USA (Member ID: 880157777)

THINK SPACE

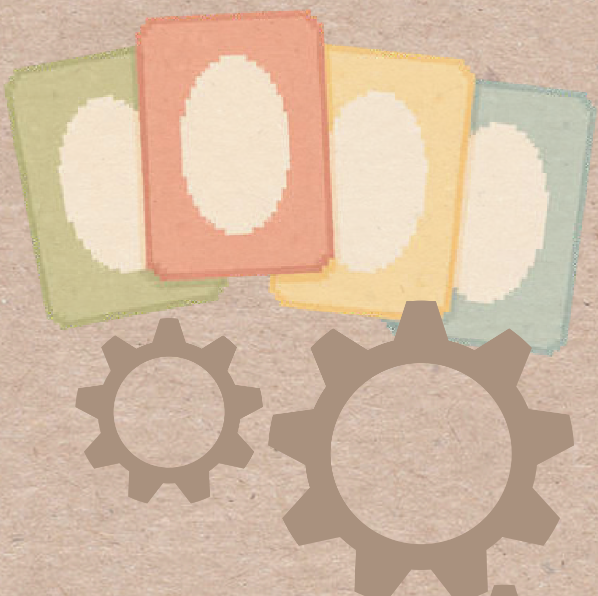
1. I balance time, cost, and quality too,
Making systems better that's what I do.
Who am I?

2. I minimize cost but maximize gain,
Finding the best path through every constraint.
What am I?

3. I ENSURE PRODUCTS MEET THE STANDARD RIGHT,
CHECKING DEFECTS BOTH DAY AND NIGHT.
WHAT AM I?

4. I improve work without adding strain,
Making tasks easier again and again.
What am I?

5. I improve quality not by inspection,
But by preventing errors before they exist.
What am I?



ANSWERS:
1. INDUSTRIAL ENGINEER 2.OPTIMIZATION 3. QUALITY CONTROL 4. ERGONOMICS 5. POKA-YOKE

RV College of Engineering®

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